

PRODUCT ENGINEERING CHALLENGE 2026

Design. Build. Present. Get Hired.

Solve real-world problems.
Build impactful solutions.
Showcase your talent.
Launch your career.



Cash Prize worth
₹1 Lakh +



Real-world
Problem Statements



Collaborate
in Teams



Build Innovative
Solutions



Present to
Industry Experts



Win Prizes &
Career Opportunities

22North Product Engineering Challenge

Design. Build. Present. Get Hired.

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Important Notes

These challenges are inspired by real enterprise customer engagements. There is intentionally no single correct solution. We are evaluating how you think, prioritise, architect and communicate – not just how many features you build.

- Understand the business problem
- Ask the right questions
- Make reasonable assumptions
- Prioritise features
- Design scalable solutions
- Build an intuitive user experience
- Deliver a working product
- Explain design decisions

Common Deliverables

- Working Product / Prototype
- Source Code
- README & Setup Instructions
- Architecture Diagram
- API Documentation (where applicable)
- Presentation Deck (Maximum 10 Slides)
- Document assumptions, trade-offs and future enhancements

Guidance for Participants

There is no expectation that every feature listed must be implemented. Focus on solving the business problem well, delivering a polished end-to-end experience, documenting assumptions and demonstrating sound engineering judgement.

Please submit your project by **Friday, 03-Jul-2026, 12:00hrs IST**.

Your submission email should be sent to: campusconnect@phonon.io

Email Subject: 22North Product Engineering Challenge 2026 | Team <Team Name> | <Project Name>

Please attach/include:

-  Source Code (GitHub repository link or ZIP file)
-  Presentation Deck (PPT/PDF)
-  Solution Document (Architecture, Design & Assumptions)
-  Demo Video (3–5 minutes) (*Optional but Recommended*)
-  README containing:
 - Team Name
 - Team Members
 - College Name
 - Build & Run Instructions
 - Technology Stack Used

Important

- Ensure all required files and links are accessible before submission.
- Submissions received after **12:00 PM IST on 03-Jul-2026** may not be considered.
- If using GitHub, make sure the repository is accessible to the evaluation team (public or shared appropriately).

What We're Looking For

- Business understanding; Product thinking
- Customer experience
- Clean architecture & Engineering quality
- Scalability and maintainability
- Clear communication

Challenges

Challenge 1 – Self-Service Flight Recovery

Customer: Regional Airline

Background

SkyJet Airways operates 65 aircraft across Asia. During weather disruptions nearly 40% of passengers call the contact center to ask whether their flight is cancelled, whether they can move to another flight, or whether they are eligible for a refund. Average wait time exceeds 25 minutes. The COO wants a digital self-service experience before the next holiday season.

Your Challenge

Design and build an MVP that helps passengers recover from disrupted journeys without contacting an agent. Decide which disruption scenarios to support, which actions to automate, and which decisions should remain agent-assisted.

Constraints

- 48-hour build window
- Passenger and flight data available through APIs
- No payment integration
- Focus on customer experience over completeness

Deliverables

- Working MVP / Prototype
- Customer Journey
- Architecture Diagram
- API Design
- Key Assumptions
- 5-minute Demonstration

Challenge 2 – Conversational Booking & Servicing

Customer: Global Airline

Background

BlueWings Airlines wants passengers to complete booking and servicing journeys through WhatsApp and Progressive Web Apps instead of calling the contact centre.

Your Challenge

Design an MVP that enables passengers to complete common booking and servicing journeys. Decide which journeys to support, how authentication should work and when to involve an agent.

Constraints

- 48-hour build window
- Booking data available via APIs
- Payments may be simulated
- Focus on customer experience

Deliverables

- Working MVP / Prototype
- Customer Journey
- Architecture Diagram
- API Design
- Key Assumptions
- 5-minute Demonstration

Challenge 3 – Investor Self-Service Platform

Customer: ABC Mutual Fund

Background

ABC Mutual Fund wants to reduce repetitive customer support requests for SIP mandate failures, statements, KYC updates and nominee changes.

Your Challenge

Design a conversational investor self-service platform and decide which journeys should be automated and where advisor intervention is required.

Constraints

- 48-hour build window
- Investor data via APIs
- No payment implementation
- Focus on rich conversations

Deliverables

- Working MVP / Prototype
- Customer Journey
- Architecture Diagram
- API Design
- Key Assumptions
- 5-minute Demonstration

Challenge 4 – AI Conversation Studio

Customer: Global Enterprise

Background

The organisation has multiple AI assistants but lacks visibility into response quality, hallucinations and prompt effectiveness.

Your Challenge

Design an AI Conversation Studio supporting knowledge management, testing, response evaluation, feedback, governance and analytics.

Constraints

- 48-hour build window
- LLMs may be mocked
- Knowledge sources may be simulated
- Focus on explainability

Deliverables

- Working MVP / Prototype
- Customer Journey
- Architecture Diagram
- API Design
- Key Assumptions
- 5-minute Demonstration

Challenge 5 – Cloud Cost Intelligence

Customer: Growing SaaS Company

Background

A rapidly growing SaaS company needs better visibility into cloud utilisation and optimisation opportunities.

Your Challenge

Design a cloud cost intelligence platform that analyses infrastructure usage and recommends actionable savings.

Constraints

- 48-hour build window
- Sample datasets may be assumed
- No live AWS integration
- Focus on actionable insights

Deliverables

- Working MVP / Prototype
- Customer Journey
- Architecture Diagram
- API Design
- Key Assumptions
- 5-minute Demonstration

Challenge 6 – Enterprise Security & Compliance

Customer: Enterprise SaaS Company

Background

The company is preparing for SOC 2 Type II certification while complying with the DPDP Act. Compliance is currently managed through spreadsheets.

Your Challenge

Design a Security & Compliance platform to manage risks, controls, audit evidence and privacy obligations.

Constraints

- 48-hour build window
- Compliance may be simplified
- Authentication may be mocked
- Focus on workflows

Deliverables

- Working MVP / Prototype
- Customer Journey
- Architecture Diagram
- API Design
- Key Assumptions
- 5-minute Demonstration

Evaluation Criteria

Judges will evaluate not only feature completeness but also simplicity, usability, scalability, engineering quality and the team's ability to justify design decisions.

The evaluation is on engineering decisions, product thinking, implementation quality & understanding - not on writing every line of code manually.

Each submission will be evaluated using the following parameters:

Criteria	Weightage
Working Product (UI/UX, Usability, Self Service etc.)	30%
Innovation & Product Thinking	20%
Solution Design and Non Functional Requirements (Security, Scalability, Performance and Deployment Infrastructure)	15%
Business Understanding	15%
Engineering Quality (Code Quality, Automation, Documentation, Testing)	10%
Presentation & Technical Demonstration	10%

Deliverables

Each participating team will submit:

- Source code
- Working demo
- Architecture diagram
- README
- API documentation
- Database schema (if applicable)
- Assumptions
- AI tools used
- 5-minute demo video
- Presentation

Awards & Career Opportunities

Cash Prizes

-  **First Prize:** ₹40,000
-  **Second Prize:** ₹25,000
-  **Third Prize:** ₹15,000
-  **Four Consolation Prizes:** ₹5,000 each

Internship & Placement Opportunities

In addition to the prize money:

- Winning teams will receive a **fast-tracked Internship Offer** or **direct interview opportunity** for full-time employment.
- Exceptional individual performers may receive internship or employment offers irrespective of their team's final ranking.
- Successful interns will be considered for **Pre-Placement Offers (PPO)** based on their performance during the internship.

Terms and conditions

- Participation is open only to eligible candidates as per the challenge eligibility criteria.
- Participants may register individually or in teams of up to two members.
- Each participant/team may submit only **one project**.
- **Original work is mandatory.** Plagiarism or unfair practices will result in immediate disqualification.
- Use of **AI tools** (e.g., ChatGPT, GitHub Copilot, Cursor, Claude, Gemini) and open-source libraries is permitted but must be disclosed.
- All submissions must be received before the specified deadline. Late submissions will not be accepted.
- Shortlisted teams must present their solution in person at the 22North office.
- The jury's decision on evaluation, awards, and shortlisting shall be final and binding.
- Winning the challenge does not guarantee an internship, PPO, or employment offer.
- 22North reserves the right to modify the schedule, rules, or evaluation process, or cancel the challenge if required.

FAQs

Who can participate?

The 22North Product Engineering Challenge is open to final and pre-final year undergraduate and postgraduate engineering students, recent graduates, and software professionals with up to two years of full-time industry experience. Participants may register individually or as a team of two members. If you register individually, we may pair you with another participant.

What kind of problems will we be solving?

The challenge is built around real enterprise product engineering problems inspired by the work we do at 22North. You will work on products across Aviation, Enterprise Communications, BFSI, Artificial Intelligence, Cloud Engineering and Security. These are not academic assignments or competitive programming questions—they are simplified versions of problems our customers face every day.

Is this primarily a coding competition?

No. While building software is an important part of the challenge, we are equally interested in understanding how you approach problems. We look for participants who can understand business requirements, ask thoughtful questions, make sensible assumptions, prioritise features, design clean solutions and communicate their ideas clearly. There is intentionally no single correct solution.

What technologies can we use?

You are free to choose the technology stack that best suits your solution. At 22North, our engineering teams primarily work with Java, Spring Boot, the MERN stack, Linux, REST APIs, PostgreSQL, MongoDB, AWS Cloud, WebRTC and modern AI platforms. However, prior experience with these technologies is not a prerequisite for participating in the challenge.

Is the work mostly frontend, backend or product-focused?

Most of our engineers work across multiple layers of the stack. Depending on the project, you may work on frontend experiences, backend services, APIs, cloud infrastructure, AI integrations and product design. We strongly believe that great

engineers understand customer problems before writing code, so product thinking is just as important as technical implementation.

Can we use AI tools during the challenge?

Absolutely. We encourage participants to use modern AI-assisted engineering tools such as ChatGPT, GitHub Copilot, Cursor, Claude, Gemini and similar platforms. Engineering today is about effectively leveraging these tools rather than avoiding them. Our evaluation focuses on your engineering judgement, design decisions and implementation quality.

Do we need to implement every feature mentioned in the project brief?

Not at all. The project briefs are intentionally broader than what can realistically be completed within the challenge duration. We expect teams to make thoughtful decisions about what to build, what to defer and how to deliver the most valuable solution within the available time. A polished, well-designed product is generally more valuable than an incomplete product with many partially implemented features.

How will submissions be evaluated?

Our evaluation goes beyond feature completion. The jury will consider how well your solution addresses the business problem, the quality of your product thinking, the user experience, engineering quality, software architecture, innovation and your ability to explain and defend your design decisions during the presentation. We are looking for future product engineers, not just good programmers.

What should we submit?

Each team should submit a working prototype along with the complete source code, setup instructions, an architecture diagram, API documentation where applicable and a presentation deck of not more than ten slides. During the final presentation, teams should also explain the assumptions they made, the trade-offs they considered and the future enhancements they would implement given additional time.

Is this primarily a hiring initiative?

Yes. The Product Engineering Challenge is our flagship talent discovery initiative. While the event includes cash prizes, our primary objective is to identify engineers who demonstrate strong product thinking, technical capability and the ability to solve real business problems. Exceptional participants may receive internship opportunities, pre-placement offers or invitations to interview for full-time roles, irrespective of whether their team wins a prize.

What internship opportunities are available?

Outstanding participants may be offered a **9-12 month Product Engineering Internship** at 22North. Interns receive a monthly stipend of **₹15,000** and work full-time from our Vadodara office. This is a structured engineering internship where participants contribute to live enterprise products under the guidance of experienced product managers, architects and engineers.

What happens after the internship?

We believe in rewarding ownership, learning agility and impact. Compensation growth at 22North is driven by performance and the ability to take on increasing product and customer responsibilities.

High-performing interns may be considered for full-time Product Engineering roles within the company. Depending on performance, skills and responsibilities, our current entry-level compensation typically ranges between **₹3.6 lakh and ₹4.8 lakh per annum for graduates** and **₹4 lakh and ₹5.4 lakh per annum for post-graduates**, depending on skills, performance and role.

Is the internship or job remote?

No. Our engineering organisation currently operates on a **100% Work from Office** model from our Vadodara headquarters. We believe that early-career engineers learn significantly faster through close collaboration, mentorship and regular design discussions with experienced colleagues.

Will shortlisted teams have to present in person?

Yes. Finalist teams will be invited to present their solutions before the evaluation panel at our Vadodara office. Apart from demonstrating your solution, this also gives

us an opportunity to interact with you, understand your thought process and discuss your engineering decisions.

Will interns work on real customer projects?

Yes. Interns and full-time engineers at 22North work on products used by leading airlines, financial institutions and enterprise customers. The challenges in this competition themselves are inspired by our actual product roadmap and customer engagements.

Does winning guarantee a job?

No. Winning a prize does not automatically result in an internship or employment offer. Similarly, outstanding participants who do not win may still receive internship or interview opportunities if they demonstrate the qualities we are looking for.

Why should I participate?

Because this is much more than a hackathon. It is an opportunity to build products that resemble real enterprise software, learn how modern product teams approach engineering problems, interact with experienced practitioners and potentially begin your career with 22North. If you enjoy building products, solving meaningful problems and learning rapidly, we believe you will find this experience rewarding.